

Topic: Hypothesis Testing

Important note for submissions:

- Only handwritten answers are allowed.
 - The hard-copy of solutions should be submitted to me in Monday lecture.
 - No email submissions will be entertained.
 - **Submit solutions to ALL Problems. Give BRIEF explanations. No more than both-sided A4-sized page should be submitted.**
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- (15 marks) You obtain a list of 4 samples y_1, y_2, y_3, y_4 from a random variable $\mathbf{Y}$ which has either the t -distribution or the F -distribution (degrees of freedom unknown). This list turns out to be 0.34, 4.36, -0.93 , 1.37. Is the random variable $\mathbf{Y}$ more likely to have the t - or F -distribution? Why?
- (15 marks) Which of the following is the best interpretation of the power of a test of hypotheses?
 - The power of the test is the proportion of times you reject H_0 ;
 - If we repeat the same test on different data sets, the power of the test is the proportion of times you reject H_0 .
 - If we repeat the same test on different data sets, the power of the test is the proportion of times you reject a false H_0 .
 - If we repeat the same test on different data sets, the power of the test is the proportion of times you reject a true H_0 .
 - The power of the test measures how false H_0 is.
- (20 marks) Using the three confidence intervals below for $(\mu_1 - \mu_2)$ based on the same data set, what is the correct range of the p -value corresponding to the same data when testing $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 \neq \mu_2$?

90% : (0.0037, 0.0963)

95% : (-0.0051, 0.1051)

99% : (-0.0225, 0.1225)

 - $p\text{-value} > 0.10$.
 - $0.05 < p\text{-value} < 0.10$.
 - $0.01 < p\text{-value} < 0.05$.
 - $p\text{-value} < 0.01$.
 - Not enough information to tell.
- (15 marks) What does the word “confidence” mean in reference to a confidence interval for the difference in population means $\mu_1 - \mu_2$?

- (a) We are confident that 0 is in the interval.
 - (b) We are confident that the difference in the sample means is in the interval.
 - (c) We are confident that the true difference in the population means lies in this particular interval.
 - (d) We are confident the procedure produces intervals containing the true difference in the population means in $(1 - \alpha)$ of repeated samples.
 - (e) We are confident that both population means are in the interval.
 - (f) We are confident that at least one of the population means is in the interval.
5. (20 marks) The average inter-pupillary distance (the distance between the eyes) for all adult males is 65 mm. A researcher studying the effects of fetal alcohol syndrome is planning to test her theory that the average inter-pupillary distance of men with fetal alcohol syndrome is greater than 65 mm. For her study she plans on selecting a random sample of 40 men with fetal alcohol syndrome. If the true average inter-pupillary distance of men with fetal alcohol syndrome is really 69 mm and the significance level is $\alpha = 0.05$, then the power of her test is 0.82. This means that in repeatedly testing on different samples from the same population,
- (a) 18% of the time they will make a type I error.
 - (b) 18% of the time they will make a type II error.
 - (c) 82% of the time they will make a type II error.
 - (d) 82% of the time the p -value will be less than 0.05.
 - (e) Two of the above are correct.
6. (15 marks) Which of the following affect the width of a confidence interval for the population proportion p ? Consider the usual large sample interval.
- (a) The sample proportion $\hat{p}$.
 - (b) The sample size n .
 - (c) The confidence coefficient $(1 - \alpha)$.
 - (d) All of the above.
 - (e) Only two of the above.